Effect of meat replacement by tofu on CHD risk factors including copper induced LDL oxidation.

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OBJECTIVE: To investigate the effect of replacing lean meat with a soy product, tofu, on coronary heart disease risk factors including serum lipoproteins, lipoprotein (a), factor VII, fibrinogen and in vitro susceptibility of LDL to oxidation. DESIGN: A randomized cross over dietary intervention study. SETTING: Free-living individuals studied at Deakin University. SUBJECTS: Forty-five free-living healthy males aged 35 to 62 years completed the dietary intervention. Three subjects were non-compliant and excluded prior to analysis. INTERVENTIONS: A diet containing 150 grams of lean meat per day was compared to a diet containing 290 grams of tofu per day in an isocaloric and isoprotein substitution. Each dietary period was one month duration. RESULTS: Analysis of the seven-day diet record showed that diets were similar in energy, protein, carbohydrate, total fat, saturated and unsaturated fat, polyunsaturated to saturated fat ratio, alcohol and fiber. Total cholesterol and triglycerides were significantly lower, and in vitro LDL oxidation lag phase was significantly longer on the tofu diet compared to the meat diet. The hemostatic factors, factor VII and fibrinogen, and lipoprotein(a) were not significantly affected by the tofu diet. CONCLUSIONS: The increase in LDL oxidation lag phase would be expected to be associated with a decrease in coronary heart disease risk.